Band-1 Carreras AC Length-14 Certificate Verify

Verification / lit-certificate replay only (NOT AC solved; AK(3) OPEN; GATE-04 LOCKED)

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Document class. Scientific computational / literature-certificate verification increment (Band-1).
Program. Open-mathematics collapse list · Human+AI attack order · ZoraASI mission-control.
Parents. `2026-09-11_Band1_Andrews_Curtis_presentation_probe`; `2026-09-14_Band1_Andrews_Curtis_AK2`
Lab. `/workspace/experiments/band1/carreras_ac_length14_verify/` (upstream mirror of `joe-carr-data/ac-certificates`).
Cutoff. 2026-09-14.

Honesty stamp (non-negotiable).
This memo is a **verification / lit-certificate replay** only. It **does not** claim the Andrews–Curtis conjecture. It **does not** trivialize or refute **AK(3)**. Equivalence of some length-14 Miller–Schupp presentations to AK(3) is **not** a trivialization of AK(3). It **does not** stamp Clay-closed or finished-ToE. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved claims. `clay_closed = false`.

Abstract

We box-replay the five public machine-checkable Andrews–Curtis **equivalence** certificates of Carreras (arXiv:2607.23611; artifact `joe-carr-data/ac-certificates`, Zenodo DOI 10.5281/zenodo.21499081) using the author's dependency-free verifier `ac_verify.py`. All five certificates **PASS** on this box (`selftest_ac.py` PASS; 5/5 certs PASS in < 1 s). Headline: the **MS(3)** branch of the length-14 collapse is **unconditionally** AC-equivalent to **AK(3)** (Thm 3/3'/4); the two remaining **MS(2)** open classes are σ-paired (Thm 1–2) but **not** collapsed to AK(3). **AK(3)** and the full AC conjecture remain **OPEN**.

Keywords. Andrews–Curtis; Akbulut–Kirby AK(3); Miller–Schupp; machine-checkable certificate; Band-1; non-claim.

1. Statement lock

Andrews–Curtis (rank 2). A balanced presentation $\langle x, y \mid r, s\rangle$ of the trivial group is AC-trivializable if a finite sequence of elementary AC moves (multiply one relator by another/inverse; invert a relator; conjugate a relator by a free generator) reaches $\langle x, y \mid x, y\rangle$ (up to the usual free-group word reductions).

Finite lemma (this increment only). Public length-14 equivalence certificates from Carreras 2026 replay with PASS on this box. **Not** AC; **not** AK(3) trivialized.

2. What changed vs prior Band-1 AC probes

Axis	2026-09-11 presentation probe	2026-09-14 AK(2) replay	This increment
Toy / AK(2)	toy trivialized; AK(2) shallow miss	box BFS 17 + Havas 14 PASS	retained
Length-14 frontier	lit ledger only	next-wave item open	**5/5 Carreras certs box-PASS**
MS(3) $\sim$ AK(3)	stated in lit, no house replay	—	**Thm3/3'/4 PASS**
MS(2) open pair	open	—	**Thm1–2 PASS** (σ-pair; still open vs AK(3))
Claim level	methods	verification replay	**same; AC OPEN; AK(3) OPEN**

3. Methods

1. Shallow-clone `https://github.com/joe-carr-data/ac-certificates` at commit `d347fc4764ee55de6fc779fe697769127dde48e3` (2026-07-22 FINAL pin + Zenodo DOI).
2. Run `python3 selftest_ac.py` (honest scramble trivialization + equivalence-mode self-tests + accounting regression).
3. Run `python3 ac_verify.py` on each theorem certificate listed in the artifact README.
4. Record SHA256 of certs + verifier; write `results.json` / `SUMMARY.txt` / this memo.
5. **No** search for new AC sequences; **no** claim beyond re-verification.

4. Results (box)

Theorem	Certificate file	Packaged [classical]	Box
Thm 1	`cert_sym_equiv_orphan1.json`	36 [58]	**PASS**
Thm 2	`cert_idx0_equiv_mirror.json`	85 [155]	**PASS**
Thm 3	`bottleneck_ak_f3_cert.json`	66 [101]	**PASS**
Thm 3	`cert_ms3_yx2y_equiv_ak3.json`	71 [106]	**PASS**
Thm 4	`cert_ms3_yinvx2yinv_equiv_ak3.json`	13 [22]	**PASS**

`selftest_ac.py`: **PASS**. Wall-clock for the five certs: sub-second on this box.

Certified connected blocks (Carreras + box replay):

- $\{\mathrm{MS}(3, yx^2y),\ \mathrm{MS}(3, y^{-1}x^2y^{-1}),\ \mathrm{AK}(3)\}$
- $\{P_1,\ \mathrm{MS}(2, yx^2y^{-1}x^{-2})\}$
- $\{\mathrm{MS}(2, yx^2yx^{-2}),\ \mathrm{MS}(2, x^{-2}y^{-1}x^2y^{-1})\}$

5. What remains OPEN

1. **AK(3)** ordinary AC-trivialization — OPEN.

2. Collapse of the two **MS(2)** open classes to AK(3) — OPEN (Thm 1–2 only realize σ within those classes).

3. Full **Andrews–Curtis** conjecture — OPEN.

4. Rank-2 length ≥ 15 frontier beyond the length-14 MS six-case benchmark — out of scope here.

6. Honesty / non-claims

- Does **not** solve Andrews–Curtis.
- Does **not** trivialize or refute AK(3).
- Equivalence certificates $\neq$ trivialization certificates.
- No Clay stamp; Forced-NS H-4.12 remains OPEN elsewhere.
- GATE-04 collection **LOCKED** (no invented J).
- No RH / P$\neq$NP / YM / finished-ToE claims.
- `clay_closed = false`.

7. Artifacts

- Lab: `/workspace/experiments/band1/carreras_ac_length14_verify/`
- Upstream pin: commit `d347fc4…` · DOI `10.5281/zenodo.21499081` · arXiv `2607.23611`
- `results.json`, `sha256sums.txt`, `verify_out.txt`, `selftest_out.txt`, `SUMMARY.txt`
- PDF twin under `/workspace/DOWNLOAD_PDFs/` and Mac `~/Downloads/DOWNLOAD_PDFs_2026-09-14/`

8. Next optional

Cap-set $a(7)$ lower-bound witness re-cert OR Hadamard remaining-$4k$ / order-2004 lit watch OR AK(3)-directed bounded search (honest non-finding only).